



Rajiv Gandhi Institute of Petroleum Technology (RGIPT)

Lab Assignment-II
Course: AIML DL Application (CS361)

Problem Statement

Porter is an intra-city logistics platform that connects customers, restaurants/businesses, and delivery partners. Accurate estimation of delivery time is important for:

1. Improving customer experience
2. Optimizing delivery-partner allocation
3. Identifying operational bottlenecks
4. Reducing delayed deliveries
5. Improving resource utilization

About Dataset:

Dataset located at -

https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/015/039/original/dataset.csv.zip?1663710760

Each row in this file corresponds to one unique delivery. Each column corresponds to a feature as explained below.

1. market_id : integer id for the market where the restaurant lies
2. created_at : the timestamp at which the order was placed
3. store id
4. actual_delivery_time : the timestamp when the order was delivered
5. store_primary_category : category for the restaurant
6. order_protocol : integer code value for order protocol(how the order was placed ie: through porter, call to restaurant, pre booked, third part etc)
7. total_items subtotal : final price of the order
8. num_distinct_items : the number of distinct items in the order
9. min_item_price : price of the cheapest item in the order
10. max_item_price : price of the costliest item in order
11. total_onshift_partners : number of delivery partners on duty at the time order was placed
12. total_busy_partners : number of delivery partners attending to other tasks
13. total_outstanding_orders : total number of orders to be fulfilled at the moment

Task:

Your task is to develop an end-to-end AI driven solution to predict delivery time in minutes using a Neural Network-based regression model and deployed trained model using any web-based application.